

Instantiating: Making Objects from Your Class

new keyword in `main()` triggers the class constructor to build a live object in your computer's memory. instantiate and configure a object cleanly in a single line.

Dot Notation (.): The syntax syntax operator used to dig inside an object to access its variables or trigger its methods (e.g., `p1.name`).

```
// This automatically jumps to the constructor code above!  
Car myCar = new Car("Red", 60);  
myCar.displayStatus(); // Prints: The Red car is going 60 mph.
```

What Happens in Computer Memory During Instantiation?

When Java executes the line `Car genericCar = new Car("Red", 55);`, it performs three distinct steps under the hood:

1. **new Car(...):** Java allocates a brand-new chunk of blank memory big enough to hold a Car object.
2. **Car("Red", 55):** The constructor function wakes up, takes your values, and writes "Red" into the color slot and 55 into the speed slot inside that fresh memory space.
3. **Car genericCar = ...:** Java assigns a variable pointer named `genericCar` to reference (point directly to) that specific spot in memory so you can find it later using dot notation.